

Reference: MNS5

PROJECT TITLE: Design, build and testing of a 3D hologram-like RGB LED fan display	STUDENT (Mechatronics): Teagan Reddy  (26012243)
CAN PROJECT BE COMPLETED IN THE 1st SEMESTER? No	
DIRECTION: Mechanical: No Mechatronic: Yes	
DESCRIPTION: Some claim that in the Venn diagram of art and engineering, there is little to no overlap. However, several artistic inventions have been the result of the synergy between the technical prowess of engineering and the sensory enjoyment associated with the arts. One example of this is the development of 3D “holographic” RGB LED fan displays. These fan displays are not truly holographic but do give a holographic effect and have captivated audiences for decades, offering a mesmerising visual experience by allowing the 3D rendering of objects in a physical space. However, traditional holographic displays are often complex and expensive to produce. This project aims to design, build, and test a 3D hologram-like RGB LED fan display that is both affordable and versatile. By leveraging the power of RGB LEDs and the principles of optical illusion, this project will create a stunning visual display that can be used for various applications, such as university open days, entertainment, and education. The LED fan display should be safe, portable and relatively low-cost. Furthermore, an app or software with a user-friendly interface should be developed to allow for the operation of the fan as well as for uploading custom designs for display. Existing software may not be used “as is” in this project – the custom software development should be integrated within the design. Consideration should be given to multi-facet design aspects such as the integration of multiple fans to allow for a larger, coordinated display unit. The design should incorporate additive manufacturing parts and should be visually appealing. Project objectives include: 1. Formulation of specifications for a 3D hologram-like RGB LED fan display unit 2. Generation and evaluation of concepts 3. Selection of suitable materials and components for manufacturing of the final design 4. User-interface/software development and integration 5. Testing of the unit using several visual display inputs of varying complexity (including static and dynamic scenes). Electronic design, soldering and programming skills will be very handy in this project.	
Elective module required (only for Mechanical): None	